

Clone PostgreSQL Cluster from One Host to Another Using pg_basebackup (No Archive Mode)

How To Backup and Restore PostgreSQL DB Cluster from One Host to Another Using pg_basebackup (No Archive Mode)

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In PostgreSQL, a cluster is a group of databases managed by one PostgreSQL server. This is just a standalone server, not like an Oracle RAC cluster.

1. Goal

- 1. Perform a consistent backup using pg_basebackup from lxicbpgdsgv01.
- 2. Restore the backup on lxicbpgdsgv02 without applying archived WALs.

2. Environment (Source / Target)

ASPECT	SOURCE	TARGET	DIFFERENCE
Hostname	lxicbpgdsgv01	lxicbpgdsgv02	Different hostnames
IP Address	192.168.2.51	192.168.2.52	Different IPs
OS	RHEL 9	RHEL 9	Same
DB Version	PostgreSQL v17.6	PostgreSQL v17.6	Same
Archive mode	No Archivelog	No Archivelog	Same
PGDATA	/pgData/pgsql17/data	/pgdata/pgsql17/data	Different path case (D vs d)
WAL Directory	/pgWal/pgsql17/wal	/pgwal/pgsql17/wal	Different path case (W vs w)
Tablespace	pg_default	pg_default	Same
Databases	DELL, ORCL	No Databases	Need to clone

3. Verify Existing Setup

```
[postgres@lxicbpgdsgv01 ~]$ psql
psql (17.6)
Type "help" for help.

postgres=# \l+

List of databases
   Name | Owner | Encoding | Locale Provider | Collate | Ctype | Locale | ICU
Rules | Access privileges | Size | Tablespace | Description
-----+-----+-----+-----+-----+-----+-----+-----
--+
```

```

dell      | postgres | UTF8      | libc          | en_SG.UTF-8 | en_SG.UTF-8 |      |
|          | 7763 kB  | pg_default |               |              |              |      |
orcl      | postgres | UTF8      | libc          | en_SG.UTF-8 | en_SG.UTF-8 |      |
|          | 7907 kB  | pg_default |               |              |              |      |
postgres  | postgres | UTF8      | libc          | en_SG.UTF-8 | en_SG.UTF-8 |      |
|          | 492 MB   | pg_default | default administrative connection database
template0 | postgres | UTF8      | libc          | en_SG.UTF-8 | en_SG.UTF-8 |      |
| =c/postgres      +| 7545 kB | pg_default | unmodifiable empty database
|          |          |          |          |          |          |      |
| postgres=CTc/postgres |          |          |
template1 | postgres | UTF8      | libc          | en_SG.UTF-8 | en_SG.UTF-8 |      |
| =c/postgres      +| 7723 kB | pg_default | default template for new databases
|          |          |          |          |          |          |      |
| postgres=CTc/postgres |          |          |
(5 rows)

```

postgres=#

postgres=# \c dell

You are now connected to database "dell" as user "postgres".

dell=#

dell=# \db

List of tablespaces

Name	Owner	Location
pg_default	postgres	
pg_global	postgres	

(2 rows)

dell=#

dell=# \dn test

List of schemas

Name	Owner
test	postgres

(1 row)

dell=#

dell=# \echo :AUTOCOMMIT

on

dell=#

```

dell=# CREATE TABLE test.emp ( name TEXT, designation TEXT, project TEXT,
company TEXT);
CREATE TABLE

```

```
dell=# INSERT INTO test.emp VALUES ('Sugi', 'DBA', 'Jetstar', 'iGATE');
INSERT 0 1
dell=# INSERT INTO test.emp VALUES ('Teja', 'DBA', 'RCM', 'iGATE');
INSERT 0 1
dell=# INSERT INTO test.emp VALUES ('RAJ', 'DBA', 'RCM', 'iGATE');
INSERT 0 1
dell=#
```

```
dell=# select * from test.emp;
```

```
  name | designation | project | company
-----+-----+-----+-----
  Sugi | DBA         | Jetstar | iGATE
  Teja | DBA         | RCM     | iGATE
  RAJ  | DBA         | RCM     | iGATE
(3 rows)
```

```
dell=#
```

```
dell=# \c orcl
```

You are now connected to database "orcl" as user "postgres".

```
orcl=#
```

```
orcl=# \dt
```

```
           List of relations
 Schema |   Name   | Type  | Owner
-----+-----+-----+-----
 public | sample_data | table | postgres
(1 row)
```

```
orcl=#
```

```
orcl=# select count(*) from sample_data;
```

```
 count
-----
  1000  <-----
(1 row)
```

```
orcl=#
```

4. Verify postgresql.conf

```
postgres=# SHOW wal_level;
```

```
wal_level
-----
 replica  <----- Should be replica
(1 row)

postgres=#
postgres=# SHOW max_wal_senders;
 max_wal_senders
-----
 10 <-----
(1 row)

postgres=# SHOW archive_mode;
 archive_mode
-----
 off  <----- No archive log mode
(1 row)

postgres=# show archive_command;
 archive_command
-----
 (disabled)
(1 row)

postgres=#
```

5. Verify pg_hba.conf

```
# TYPE      DATABASE      USER      ADDRESS      METHOD
# Local connections for replication (for pg_basebackup run locally)
local      replication    all                               trust

# Remote connections for replication (for pg_basebackup run remotely)
#host      replication    repl_user    192.168.2.52/32    scram-sha-256
```

6. Verify user permissions:REPLICATION or SUPERUSER required

```
postgres=# \du
List of roles
Role name | Attributes
-----+-----
```

postgres | **Superuser**, Create role, Create DB, Replication, Bypass RLS

postgres=#

7. Directory Permissions for Backup

```
[postgres@lxicbpgdsgv01 ~]$ ls -ld /pgBackup/pgsql17/backup/
drwx-----. 2 postgres postgres 6 Oct  9 18:46 /pgBackup/pgsql17/backup/
[postgres@lxicbpgdsgv01 ~]$
```

8. Take Backup

- Want **symlinks preserved** → use **(both -Fp & -waldir)-Fp --waldir=/pgWal/pgsql17/wal**
- Want **single tar archive** → use **-Ft**, but **recreate symlinks after restore, for pg_wal**
- The **-waldir** option in **pg_basebackup** is supported only when using the plain format **(-Fp)**, not with the tar format **(-Ft)**.

```
[postgres@lxicbpgdsgv01 ~]$ nohup pg_basebackup -U postgres -D
/pgBackup/pgsql17/backup/pg_basebackup_lxicbpgdsgv01_10sep2025 -Ft -Xs -P >
/pgBackup/pgsql17/backup/pg_basebackup_lxicbpgdsgv01_10sep2025.log 2>&1 &
[1] 4438
[postgres@lxicbpgdsgv01 ~]$
```

**** I have forgot use **-v** (verbose option to get extra output)

```
nohup pg_basebackup -U postgres -D
/pgBackup/pgsql17/backup/pg_basebackup_lxicbpgdsgv01_10sep2025 -Ft -Xs -P -
v > /pgBackup/pgsql17/backup/pg_basebackup_lxicbpgdsgv01_10sep2025.log 2>&1
&
```

9. Verify Backup

```
[postgres@lxicbpgdsgv01 ~]$ cat
/pgBackup/pgsql17/backup/pg_basebackup_lxicbpgdsgv01_10sep2025.log
nohup: ignoring input
waiting for checkpoint
133816/536026 kB (24%), 0/1 tablespace
299960/536026 kB (55%), 0/1 tablespace
447864/536026 kB (83%), 0/1 tablespace
536037/536037 kB (100%), 0/1 tablespace
536037/536037 kB (100%), 1/1 tablespace
[postgres@lxicbpgdsgv01 ~]$
```

```
[postgres@lxicbpgdsgv01 ~]$ ls -lrth
/pgBackup/pgsql17/backup/pg_basebackup_lxicbpgdsgv01_10sep2025
total 540M
-rw-----. 1 postgres postgres 524M Oct  9 18:56 base.tar
-rw-----. 1 postgres postgres 223K Oct  9 18:56 backup_manifest
-rw-----. 1 postgres postgres  17M Oct  9 18:56 pg_wal.tar
[postgres@lxicbpgdsgv01 ~]$
```

10. Transfer Backup to Target

```
[postgres@lxicbpgdsgv01 pg_basebackup_lxicbpgdsgv01_10sep2025]$ scp *
192.168.2.52:/pgbackup/pgsql17/backup/
postgres@192.168.2.52's password:
backup_manifest          100% 222KB 30.7MB/s  00:00
base.tar                 100% 523MB 74.0MB/s  00:07
pg_wal.tar               100%  16MB 43.5MB/s  00:00
[postgres@lxicbpgdsgv01 pg_basebackup_lxicbpgdsgv01_10sep2025]$
```

11. Prepare Target & Restore

```
-- List Source Backup files
[postgres@lxicbpgdsgv02 ~]$ ls -lrth /pgbackup/pgsql17/backup
total 540M
-rw-----. 1 postgres postgres 223K Oct  9 19:08 backup_manifest
-rw-----. 1 postgres postgres 524M Oct  9 19:08 base.tar
-rw-----. 1 postgres postgres  17M Oct  9 19:08 pg_wal.tar
[postgres@lxicbpgdsgv02 ~]$
```

11.1 Backup Target (Optional)

```
nohup pg_basebackup -U postgres -D
/pgbackup/pgsql17/pg_basebackup_lxicbpgdsgv02_10sep2025 -Ft -Xs -P >
/pgbackup/pgsql17/pg_basebackup_lxicbpgdsgv02_10sep2025.log 2>&1 &
```

```
-- Verify Tablespace path -- It's pg_default in our case
-- Verify WAL File (Redo log files) location
```

```
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/pg_wal
```

```
lrwxrwxrwx. 1 postgres postgres 18 Oct  9 17:37 /pgdata/pgsql17/data/pg_wal
-> /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$
```

11.2 Stop PostgreSQL and Clean Directories

```
-- Stop PostgreSQL DB Cluster
[root@lxicbpgdsgv02 ~]# systemctl stop postgresql-17.service
[root@lxicbpgdsgv02 ~]#
[root@lxicbpgdsgv02 ~]# ps -ef | grep postgres
root      10259    7832  0 19:20 pts/0    00:00:00 grep --color=auto
postgres
[root@lxicbpgdsgv02 ~]#

-- Remove all from Data directory

[postgres@lxicbpgdsgv02 ~]$ rm -rf /pgdata/pgsql17/data/*
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/
total 0
[postgres@lxicbpgdsgv02 ~]$

-- Remove all from WAL directory

[postgres@lxicbpgdsgv02 ~]$ rm -rf /pgwal/pgsql17/wal/*
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgwal/pgsql17/wal/
total 0
[postgres@lxicbpgdsgv02 ~]$
```

11.3 Restore Data and WAL

```
-- List source backup files

[postgres@lxicbpgdsgv02 ~]$ cd /pgbackup/pgsql17/backup/
[postgres@lxicbpgdsgv02 backup]$ ls -ltr
total 552652
-rw-----. 1 postgres postgres 227724 Oct  9 19:08 backup_manifest
-rw-----. 1 postgres postgres 548901888 Oct  9 19:08 base.tar
-rw-----. 1 postgres postgres 16778752 Oct  9 19:08 pg_wal.tar
drwxr-xr-x. 3 postgres postgres 18 Oct  9 19:32
data_lxicbpgdsgv02_old
[postgres@lxicbpgdsgv02 backup]$
```


Restore DATA Directory

```
[postgres@lxicbpgdsgv02 ~]$ nohup tar -xvf
/pgbackup/pgsql17/backup/base.tar -C /pgdata/pgsql17/data >
/pgbackup/pgsql17/backup/base_restore.log 2>&1 &
[1] 11289
[postgres@lxicbpgdsgv02 ~]$
```

Restore WAL Directory

```
[postgres@lxicbpgdsgv02 ~]$ nohup tar -xvf
/pgbackup/pgsql17/backup/pg_wal.tar -C /pgwal/pgsql17/wal >
/pgbackup/pgsql17/backup/pg_wal_restore.log 2>&1 &
[1] 11304
[postgres@lxicbpgdsgv02 ~]$
```

-- OR --

Restore base and WAL archives sequentially in a single command

```
nohup bash -c "tar -xvf /pgbackup/pgsql17/backup/base.tar -C
/pgdata/pgsql17/data && tar -xvf /pgbackup/pgsql17/backup/pg_wal.tar -C
/pgwal/pgsql17/wal" > /pgbackup/pgsql17/backup/fulltar_restore.log 2>&1 &
```

11.4 Set Permissions

```
[postgres@lxicbpgdsgv02 ~]$ chown -R postgres:postgres /pgdata/pgsql17/data
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgdata/pgsql17/data
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ chown -R postgres:postgres /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$
```

11.5 Verify Restored Files

```
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data
total 72
-rw-----. 1 postgres postgres 88 Sep 30 21:50 postgresql.auto.conf
drwx-----. 2 postgres postgres 18 Sep 30 21:50 pg_xact
-rw-----. 1 postgres postgres 3 Sep 30 21:50 PG_VERSION
drwx-----. 2 postgres postgres 6 Sep 30 21:50 pg_twophase
drwx-----. 2 postgres postgres 6 Sep 30 21:50 pg_tblspc
```

```

drwx-----. 2 postgres postgres      6 Sep 30 21:50 pg_subtrans
drwx-----. 2 postgres postgres      6 Sep 30 21:50 pg_stat_tmp
drwx-----. 2 postgres postgres      6 Sep 30 21:50 pg_snapshots
drwx-----. 2 postgres postgres      6 Sep 30 21:50 pg_serial
drwx-----. 2 postgres postgres      6 Sep 30 21:50 pg_notify
drwx-----. 4 postgres postgres     36 Sep 30 21:50 pg_multixact
-rw-----. 1 postgres postgres 2640 Sep 30 21:50 pg_ident.conf
drwx-----. 2 postgres postgres      6 Sep 30 21:50 pg_dynshmem
drwx-----. 2 postgres postgres      6 Sep 30 21:50 pg_commit_ts
-rw-----. 1 postgres postgres 30702 Oct  7 18:20 postgresql.conf.bkp
drwx-----. 2 postgres postgres      84 Oct  9 00:01 log
-rw-----. 1 postgres postgres 1169 Oct  9 16:31
postgresql.conf.bkp_10sep2025
-rw-----. 1 postgres postgres 1171 Oct  9 16:32 postgresql.conf
drwx-----. 7 postgres postgres      59 Oct  9 18:08 base
drwx-----. 2 postgres postgres      6 Oct  9 18:44 pg_stat
-rw-----. 1 postgres postgres      30 Oct  9 18:44 current_logfiles
-rw-----. 1 postgres postgres 5600 Oct  9 18:53 pg_hba.conf
-rw-----. 1 postgres postgres      0 Oct  9 18:56 tablespace_map
drwx-----. 2 postgres postgres      6 Oct  9 18:56 pg_replslot
drwx-----. 4 postgres postgres      68 Oct  9 18:56 pg_logical
-rw-----. 1 postgres postgres 227 Oct  9 18:56 backup_label
drwx-----. 4 postgres postgres 45 Oct  9 20:06 pg_wal <--- Created as
Directory, instead of symbolic link
drwx-----. 2 postgres postgres 4096 Oct  9 20:06 global
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgwal/pgsql17/wal
total 16384
-rw-----. 1 postgres postgres 16777216 Oct  9 18:56
00000001000000000000000045
[postgres@lxicbpgdsgv02 ~]$

```

Please note:

```

--- Want symlinks preserved → use -Fp
--- Want single tar archive → use -Ft, but recreate symlinks after restore,
for pg_wal

```

11.6 Create Symbolic Link for WAL

```

[postgres@lxicbpgdsgv02 ~]$ ls -ld /pgdata/pgsql17/data/pg_wal
drwx-----. 4 postgres postgres 45 Oct  9 20:06 /pgdata/pgsql17/data/pg_wal
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ rm -rf /pgdata/pgsql17/data/pg_wal

```

```
[postgres@lxicbpgdsgv02 ~]$ ln -s /pgwal/pgsql17/wal
/pgdata/pgsql17/data/pg_wal
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ ls -ld /pgdata/pgsql17/data/pg_wal
lrwxrwxrwx. 1 postgres postgres 18 Oct  9 21:04 /pgdata/pgsql17/data/pg_wal
-> /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ cd /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 wal]$ ll
total 16384
-rw-----. 1 postgres postgres 16777216 Oct  9 18:56
00000001000000000000000045
[postgres@lxicbpgdsgv02 wal]$
```

12. Start PostgreSQL on Target — no recovery needed.

```
[root@lxicbpgdsgv02 ~]# systemctl start postgresql-17.service
[root@lxicbpgdsgv02 ~]# systemctl status postgresql-17.service
• postgresql-17.service - PostgreSQL 17 database server
   Loaded: loaded (/usr/lib/systemd/system/postgresql-17.service;
   enabled; preset: disabled)
   Active: active (running) since Thu 2025-10-09 21:06:08 +08; 5s ago
     Docs: https://www.postgresql.org/docs/17/static/
   Process: 12375 ExecStartPre=/usr/pgsql-17/bin/postgresql-17-check-db-
   dir ${PGDATA} (code=exited, status=0/SUCCESS)
   Main PID: 12380 (postgres)
     Tasks: 7 (limit: 20496)
    Memory: 33.6M
       CPU: 114ms
   CGroup: /system.slice/postgresql-17.service
           └─12380 /usr/pgsql-17/bin/postgres -D /pgdata/pgsql17/data
           └─12381 "postgres: logger "
           └─12382 "postgres: checkpointer "
           └─12383 "postgres: background writer "
           └─12385 "postgres: walwriter "
           └─12386 "postgres: autovacuum launcher "
           └─12387 "postgres: logical replication launcher "
```

Oct 09 21:06:06 lxicbpgdsgv02.rajasekhar.com systemd[1]: Starting PostgreSQL 17 database server...

Oct 09 21:06:06 lxicbpgdsgv02.rajasekhar.com postgres[12380]: 2025-10-09 21:06:06.781 +08 [12380] LOG: redirecting log output to logging collector process

Oct 09 21:06:06 lxicbpgdsgv02.rajasekhar.com postgres[12380]: 2025-10-09 21:06:06.781 +08 [12380] HINT: Future log output will appear in directory

```
"log".
```

```
Oct 09 21:06:08 lxicbpgdsgv02.rajasekhar.com systemd[1]: Started PostgreSQL  
17 database server.
```

```
[root@lxicbpgdsgv02 ~]#
```

13. Final Verification

```
[postgres@lxicbpgdsgv02 ~]$ psql
```

```
psql (17.6)
```

```
Type "help" for help.
```

```
postgres=# \l+
```

```
List of databases
```

Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU
Rules	Access privileges	Size	Tablespace	Description			
dell	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8		
		7609 kB	pg_default				
orcl	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8		
		7753 kB	pg_default				
postgres	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8		
		492 MB	pg_default	default administrative connection database			
template0	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8		
=c/postgres		7545 kB	pg_default	unmodifiable empty database			
postgres=CTc/postgres							
template1	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8		
=c/postgres		7569 kB	pg_default	default template for new databases			
postgres=CTc/postgres							

(5 rows)

```
postgres=# \db
```

```
List of tablespaces
```

Name	Owner	Location
pg_default	postgres	
pg_global	postgres	

(2 rows)

```
postgres=#
```

```
postgres=# \c dell
```

```
You are now connected to database "dell" as user "postgres".
```

```
dell=# \dt test.*
```

```
      List of relations
```

```
 Schema | Name | Type | Owner
```

```
-----+-----+-----+-----
```

```
 test   | emp   | table | postgres
```

```
(1 row)
```

```
dell=#
```

```
dell=# select * from test.emp; <----- We can see Data
```

```
 name | designation | project | company
```

```
-----+-----+-----+-----
```

```
 Sugi | DBA          | Jetstar | iGATE
```

```
 Teja | DBA          | RCM     | iGATE
```

```
 RAJ  | DBA          | RCM     | iGATE
```

```
(3 rows)
```

```
dell=#
```

```
dell=# \c orcl
```

```
You are now connected to database "orcl" as user "postgres".
```

```
orcl=# \dt
```

```
      List of relations
```

```
 Schema | Name       | Type | Owner
```

```
-----+-----+-----+-----
```

```
 public | sample_data | table | postgres
```

```
(1 row)
```

```
orcl=#
```

```
orcl=# select count(*) from sample_data;
```

```
 count
```

```
-----
```

```
 1000 <-----
```

```
(1 row)
```

```
orcl=#
```

Caution: Your use of any information or materials on this website is entirely at your own risk. It is provided for educational purposes only. It has been tested internally, however, we do not guarantee that it will work for you. Ensure that you run it in your test environment before using.

Thank you,

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